Analysis of the Airport Network of India

as a Complex Weighted Network

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Central Focus: This paper studies India's domestic airport network (ANI) as a complex network, analyzing both its *topological structure* (unweighted network) and *traffic dynamics* (weighted network). The study reveals how national transportation infrastructure differs from global networks and provides insights into network formation mechanisms shaped by geo-political constraints.

Context & Significance

Transportation infrastructure is a critical indicator of economic growth. The civil aviation sector in India has been developing steadily, accelerated by policy shifts and the addition of low-cost private carriers. Understanding these transportation systems is important for policy, administration, and efficiency.

Complex network analysis has been applied to various transportation systems (railways, airlines) from different perspectives. Prior studies include:

- **World-wide Airport Network (WAN)** — studied for topology and traffic dynamics
- **Airport Network of China (ANC)** — analyzed for small-world features and two-regime power-law distribution

ANI provides a case study of a *national* airport network, whose formation mechanism differs from global networks.

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Unweighted vs. Weighted Networks

Unweighted Network (Binary Network) — A network representation that considers only the *presence* or *absence* of a connection between nodes, ignoring the intensity or strength of interactions. Represents the **architecture** (topology) of connectivity.

For ANI: An unweighted network where nodes are airports and a link exists if there is *any* flight between two airports (regardless of frequency).

Weighted Network — A network representation that considers the *strength* or *intensity* of interactions between nodes, captured by numerical weights on edges. Represents **traffic dynamics** in the network.

For ANI: A weighted network where the weight of a link is the *number of flights per week* between two airports.

Why both representations matter: The unweighted network reveals structural properties (who connects to whom), while the weighted network reveals functional properties (how much traffic flows where). Many properties differ significantly between the two.

Matrix Representations

Adjacency Matrix (A) — An $N \times N$ binary matrix representing an unweighted network, where:

- $a_{ij} = 1$ if there is a flight from airport i to airport j (on any day of the week, by any service provider)
- $a_{ij} = 0$ if there is no flight connection

For directed networks, A is asymmetric. For undirected networks (or symmetrized networks), A is symmetric: $a_{ij} = a_{ji}$.

Weight Matrix (W) — An $N \times N$ matrix representing a weighted network, where each element w_{ij} contains the *weight* (strength) of the connection from node i to node j.

For ANI: w_{ij} = total number of flights per week from airport i to airport j.

The weight matrix is symmetrized for ANI because most routes are bidirectional (flights go both ways with equal frequency).

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Data Source

The network was constructed from a timetable of air services within India (as of January 12, 2004). Includes domestic services by all major providers:

- Indian Airlines, Alliance Air, Jet Airways, Air Sahara, Air Deccan, Jagson
- Some international airlines on domestic routes: Druk Air, Air India, Bangladesh Biman, Royal Nepal, Srilankan Airlines

Network Size & Characteristics

ANI Network Statistics:

- **Nodes (N):** 79 domestic airports
- **Directed links:** 442 flights (directed connections)
- **Undirected edges (M):** 228 flight routes (after symmetrization)
- **Bidirectional routes:** 221 out of 228 routes (97%)
- **Average degree:** $\langle k \rangle = 2M/N = 5.77$

- **Network type:** Directed, but nearly symmetric (only 14 fictitious flights added to symmetrize)

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Degree & Degree Distribution

Degree (k_i) — The number of direct connections a node has to other nodes. For airport i, the degree is the number of airports it is directly connected to by flights.

In-degree (k_i^{in}) and Out-degree (k_i^{out}) — In a directed network:

- **In-degree:** Number of incoming links (flights terminating at airport i)
- **Out-degree:** Number of outgoing links (flights originating from airport i)
- **Total degree:** $k_i = k_i^{in} + k_i^{out}$

Observation in ANI: For most airports, $k_i^{in} = k_i^{out}$ (symmetric traffic).

Degree Distribution ($P(k)$) — The probability distribution of node degrees across the network. Reveals the topology and formation mechanism.

- **Poisson distribution:** Characteristic of random networks (most nodes have similar degree)
- **Power-law distribution:** $P(k) \sim k^{-\gamma}$ — characteristic of scale-free networks (wide variation in degrees, presence of hubs)

Cumulative Degree Distribution ($P(> k)$) — The probability that a randomly selected node has degree *greater than* k. Often used for small networks to reduce statistical noise. Scaling exponent: $\gamma_{cum} = \gamma - 1$.

ANI Degree Distribution Finding:
ANI exhibits a **truncated power-law** degree distribution with scaling exponent $\gamma = 2.2 \pm 0.1$. The power-law holds over a wide range but deviates (truncates) for very large degrees due to:

- Cost of adding links to nodes
- Limited capacity of nodes (airports have physical and operational limits)

This is characteristic of *constrained* scale-free networks.

Small-World Properties

Shortest Path Length (L_{ij}) — The minimum number of flights (links) needed to travel from airport i to airport j. Also called the *distance* between nodes.

Average Shortest Path Length (L) — The mean of shortest path lengths over all pairs of nodes in the network. Measures the typical separation between two airports.

For a directed network: $L = \frac{1}{N(N-1)} \sum_{i,j=1,i \neq j}^{N} L_{ij}$

Clustering Coefficient (C_i) — The fraction of a node's neighbours that are also neighbours of each other. Measures local connectivity density.

- $C_i = 0$: None of node i's neighbours are connected
- $C_i = 1$: All of node i's neighbours form a complete graph (clique)
- Probabilistic interpretation: C_i is the probability that two neighbours of node i are also connected

Average clustering coefficient: $C = \frac{1}{N} \sum_{i=1}^{N} C_i$

Small-World Network — A network characterized by:

1. **Short average path length:** $L \sim L_{random} \approx \ln(N)/\ln(\langle k \rangle)$
2. **High clustering coefficient:** $C \gg C_{random} \approx \langle k \rangle / N$

Small-world networks combine local clustering (like regular lattices) with global efficiency (like random graphs).

ANI Small-World Properties:

- **Average shortest path length:** $L = 2.2593$
  (comparable to random network: $L_{random} \sim 2.493$)
- **Average clustering coefficient:** $C = 0.6574$
  (order of magnitude higher than random: $C_{random} \sim 0.0731$)
- **Conclusion:** ANI is a small-world network

Comparison: Airport Network of China (ANC) has similar properties: $L = 2.067$, $C = 0.733$.

Network Diameter & Path Analysis

Network Diameter — The longest shortest path in the network; the maximum distance between any two nodes.

ANI diameter: 4 — meaning at most 3 flight changes are needed to reach any airport from any other airport.

Shortest Path Distribution — The frequency distribution of all shortest paths in the network.

ANI has 6,162 distinct paths (node-to-node routes):

- Path length 1 (direct flights): 442 routes (7.2%) — 0 flight changes
- Path length 2: 3,741 routes (60.7%) — 1 flight change
- Path length 3: 1,918 routes (31.1%) — 2 flight changes

- Path length 4: 61 routes (1.0%) — 3 flight changes

⇒ About **99% of routes** are reachable with at most 2 flight changes.
⇒ About **68% of routes** are reachable with at most 1 flight change.

Interpretation: The network has evolved to balance passenger convenience (few flight changes) with economic viability for service providers. The small-world topology reflects this optimization.

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Strength & Centrality Measures

Strength (Node Strength, s_i) — The weighted counterpart of degree. The total traffic handled by an airport per week, defined as the sum of weights of all links connected to node i:
$s_i = \sum_{j=1}^{N} a_{ij} w_{ij}$
Interpretation: Strength measures the *importance* of a node in terms of traffic volume, not just the number of connections.

Correlation Between Degree and Strength — The relationship $s(k) \sim k^{\beta}$ describes how average strength scales with degree:

- $\beta = 1$: Strength and degree are uncorrelated (weights provide no new information)
- $\beta > 1$: Larger nodes handle disproportionately more traffic (strong correlation)

ANI Strength-Degree Relationship:
$s(k) \sim k^{1.43 \pm 0.06}$ (i.e., $\beta = 1.43$)
Interpretation: Strength is strongly correlated with degree. Larger airports (higher degree) handle disproportionately more traffic than expected from their connectivity alone. This is similar to WAN, where $\beta_{WAN} = 1.5 \pm 0.1$.

Weight Distribution

Weight Distribution ($P(w)$) — The probability distribution of link weights (number of flights per week on each route). Shows how traffic is distributed across routes.

ANI finding: The weight distribution is *right-skewed*, indicating high heterogeneity — a few routes carry very heavy traffic, while most carry relatively little. Similar to WAN and ANC.

Implication: Individual weights alone don't capture network complexity. Aggregate measures like strength and weighted clustering are needed.

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Weighted Clustering Coefficient

Weighted Clustering Coefficient (c_i^w) — A modified clustering coefficient that accounts for link weights (traffic intensity) on local triplets:
$c_i^w = \frac{1}{s_i(k_i-1)} \sum_{j,h} \frac{w_{ij}+w_{ih}}{2} a_{ij}a_{ih}a_{jh}$
Measures local cohesiveness by considering the interaction intensity on interconnected triplets (triangles). Normalized: $0 \leq c_i^w \leq 1$.

Average Weighted Clustering (C^w) — The mean of weighted clustering coefficients over all nodes. Indicates whether traffic is concentrated on interconnected groups.

Comparison: If $C^w/C \approx 1$, weights are uniformly distributed. If $C^w/C > 1$, traffic accumulates on interconnected triplets.

ANI Clustering Findings:

- $C^w/C \approx 1.075$ — traffic is accumulated on interconnected groups of airports (trunk routes)
- $C(k)$ shows **power-law decay**: $C(k) \sim k^{-1}$ — signature of **hierarchical structure**
- For small $k < 8$: $C(k)$ remains constant
- For larger $k \geq 8$: $C(k)$ falls rapidly — large airports (hubs) connect to distant airports that are not themselves interconnected
- $C^w(k)$ is consistently higher than $C(k)$, especially for large k — high-degree nodes form interconnected groups with high-traffic links (**rich-club phenomenon**)

Hierarchical Network — A network where the clustering coefficient decays as a power law: $C(k) \sim k^{-\alpha}$. Indicates that low-degree nodes belong to tightly interconnected communities, while high-degree nodes (hubs) connect different communities.

ANI: Shows hierarchical structure with $C(k) \sim k^{-1}$.

Trunk Routes — High-traffic corridors between major airports. In ANI, traffic is concentrated on trunk routes connecting large hubs, forming interconnected groups with heavy traffic flow.

Rich-Club Phenomenon — The tendency of high-degree nodes (hubs) to form densely interconnected cliques with other high-degree nodes. Measured by comparing $C^w(k)$ with $C(k)$ — when $C^w(k) \gg C(k)$ for large k, rich-club behavior is present.

ANI and WAN both exhibit rich-club behavior.

Degree Correlations

Average Nearest Neighbor Degree ($k_{nn}(k)$) — The average degree of the neighbors of nodes with degree k. Reveals correlations in network topology:

$k_{nn}(k) = \sum_{k'} k' P(k'|k)$

where $P(k'|k)$ is the conditional probability that a node of degree k connects to a node of degree k'.

- If $k_{nn}(k)$ is independent of k: no degree correlations (random)
- If $k_{nn}(k)$ increases with k: **assortative mixing** (hubs connect to hubs)
- If $k_{nn}(k)$ decreases with k: **disassortative mixing** (hubs connect to low-degree nodes)

Assortative Mixing — The tendency of high-degree nodes to preferentially connect to other high-degree nodes. Characteristic of social networks.

Example: In WAN, hubs (major international airports) connect to many other hubs.

Disassortative Mixing — The tendency of high-degree nodes to connect predominantly to low-degree nodes. Characteristic of many technological and biological networks.

Example: In ANI, major hubs provide connectivity to many small regional airports.

Weighted Nearest Neighbor Degree ($k^w_{nn}(k)$) — A weighted version that accounts for traffic strength when computing nearest neighbor degree:

$k^w_{nn,i} = \frac{1}{s_i} \sum_{j=1}^{N} a_{ij} w_{ij} k_j$

Measures the effective tendency to connect with high- or low-degree neighbors according to the magnitude of actual interactions (traffic).

ANI Degree Correlation Findings:

- For small degrees ($k < 8$): Uncorrelated — no clear assortative or disassortative pattern
- For large degrees ($k \geq 8$): **Clear disassortative mixing** — hubs connect to many low-degree airports
- **Global measure:** Assortativity coefficient $r = -0.4016$ (negative $\Rightarrow$ disassortative)
- **Contrast with WAN:** WAN shows assortative mixing for small k — global hubs connect to other global hubs
- $k^w_{nn}(k)$ is consistently *higher* than $k_{nn}(k)$, especially for large k — despite disassortative topology, **traffic is concentrated between high-degree nodes**

Assortativity Coefficient (r) — A global quantitative measure of degree correlations, ranging from -1 to $+1$:

- $r = 0$: No correlations (random)
- $r > 0$: Assortative mixing
- $r < 0$: Disassortative mixing

ANI: $r = -0.4016$ (disassortative)

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ANI vs. WAN — Similarities & Differences

Similarities Between ANI and WAN:

- Both are small-world networks (short L, high C)
- Both have scale-free degree distributions (truncated for large k)
- Both show strong correlation between strength and degree ($\beta > 1$)
- Both exhibit rich-club phenomenon (hubs form high-traffic cliques)
- Traffic accumulates on interconnected groups (trunk routes)

Key Difference — Degree Correlations:

- **WAN:** Assortative mixing — global hubs connect to other global hubs (international connectivity)
- **ANI:** Disassortative mixing — national/regional hubs connect to many small airports (domestic connectivity)
- **Reason:** *Geo-political constraints.* National hubs have political/economic compulsion to provide connectivity to remote, low-traffic airports. In contrast, global hubs prioritize connecting to other major international airports.

Traffic Offset: Although ANI's *topology* is disassortative, the *traffic dynamics* compensate — traffic is still concentrated between high-degree nodes (trunk routes). This is revealed by $k^w_{nn}(k) > k_{nn}(k)$.

Hierarchical Structure in ANI

Hierarchy in Networks — A structural organization where nodes are arranged in levels. Low-degree nodes form tightly connected local communities; high-degree nodes (hubs) connect different communities. Detected by power-law decay of $C(k)$.
ANI: Shows hierarchical signature with $C(k) \sim k^{-1}$

Why ANI has hierarchy but not other infrastructure networks:
Some infrastructure networks (Internet routers, power grids) constrained by geography do NOT show hierarchy because the cost of adding a link grows with distance. In airport networks, geographical constraints are weaker — it's relatively easier to add air routes than physical cables over comparable distances.

Interpretation — National vs. Global Networks

Formation Mechanism of National Airport Networks:

1. **Regional/National Hubs:** Provide connectivity to many small airports in their region or country
2. **Geo-political Constraints:** Policy requirements to connect remote or underserved regions
3. **Limited Number of Hubs:** Unlike global networks, national networks have fewer major hubs, leading to disassortative topology
4. **Traffic Concentration:** Economic incentives drive airlines to concentrate traffic on profitable trunk routes (hub-to-hub), offsetting topological disassortativity
5. **Hierarchical Organization:** Small airports cluster into regional groups connected by hubs; hubs connect to each other via trunk routes

Comparison: Airport Network of China (ANC) also shows disassortative mixing — this may be a general feature of national transportation networks.

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Property	**ANI Value / Finding**
Number of nodes (N)	79 airports
Number of edges (M)	228 routes
Average degree ($\langle k \rangle$)	5.77
Average shortest path (L)	2.2593
Clustering coefficient (C)	0.6574
Network diameter	4
Small-world?	Yes
Degree distribution	Truncated power-law, $\gamma = 2.2 \pm 0.1$
Strength-degree scaling	$s(k) \sim k^{1.43}$
Hierarchical structure?	Yes, $C(k) \sim k^{-1}$
Degree mixing	Disassortative ($r = -0.4016$)
Rich-club phenomenon?	Yes
Traffic concentration	On trunk routes (interconnected hubs)

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Main Conclusions:

1. ANI, despite being small ($N = 79$), exhibits complex dynamics similar to larger airport networks
2. It is a **small-world network** with **hierarchical structure** and **truncated scale-free** degree distribution
3. ANI has **disassortative mixing**, contrasting with WAN's assortative mixing — attributed to geo-political constraints on national infrastructure

4. Traffic is **accumulated on trunk routes** (interconnected hubs) and concentrated between large airports, offsetting topological disassortativity
5. The network has evolved to balance **passenger convenience** (few flight changes needed) with **economic viability** for airlines
6. The formation mechanism of **national transportation networks differs** from global networks due to regional connectivity requirements

Future research directions:
ANI is expected to grow rapidly with addition of airports, low-cost carriers, and increased traffic complexity. Studying its evolution offers an excellent opportunity to understand and model the topology, traffic dynamics, and geo-political constraints that shape national infrastructure of economic importance.

Summary compiled from: Bagler, G. “Analysis of the Airport Network of India as a complex weighted network.” *Phys. Rev. E* (2008) · arXiv:cond-mat/0409773v2 [cond-mat.other]